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A Runtime Anomaly Detector via Service Communication Proxy for 5G Mobile Networks

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***Abstract*—With the growing popularity of the 5G mobile network, its security is becoming important. Although the newly introduced 5G security mechanisms have addressed many legacy security issues, there may be still vulnerabilities in the 5G network due to newly deployed components and used technologies. To detect security threats, we develop a runtime anomaly detector (RAD) platform, designated as 5G-RAD, to cooperate with the operational 5G core network via the service communication proxy (SCP). It validates the core network operation in terms of state machine and message content by analyzing control-plane messages. We demonstrate its effectiveness by building a 5G mobile network architecture with SCP based on the open-source free5GC and UERANSIM. The 5G-RAD is tested with three attacks, including DoS, authentication bypass, and invalid message injection; it can successfully detect them at run time.**

***Index Terms*—5G, mobile network, security, anomaly detection**

I. Introduction

5G mobile network has been rolled out rapidly and becoming popular in recent years. To connect everything together while supporting diversified services, the 5G core network is designed with a flexible service-based architecture (SBA) based on network function virtualization (NFV). It decomposes the control-plane functionality into multiple virtualized network functions (NFs), e.g., access and mobility management function (AMF) and authentication server function (AUSF); each NF service exposes its functionality through the service-based interface (SBI). Such cloud-native SBA can enable the 5G network to support varying service needs.

To strengthen security protection, the 5G mobile network has gained security lessons learned from the legacy networks and addressed most of discovered security issues. However, it may still suffer from security threats, since adopting new technologies can increase its attack surface and each new component may have security vulnerabilities from design flaws, implementation bugs, or/and operation slips. Thus, for operational 5G mobile networks, especially the private 5G networks with low-cost components, it calls for an anomaly detection solution to detect malicious attempts and anomalous operations, thereby being able to address them timely.

We thus develop a runtime anomaly detector (RAD) platform, designated as 5G-RAD, to cooperate with the operational 5G core network in a standard-compliant way based on the service communication proxy (SCP), which is newly introduced in 3GPP Release 16 to support indirect communication [1]. While the SCP is mediating control-plane messages between NFs, all the messages can be duplicated and forwarded to the 5G-RAD at run time for the validation of the corresponding control-plane operations. In this work, two dimensions of the 5G operation context are considered for each user equipment (UE) in the validation: state machine at each NF and constraints of information elements (IEs) in each message; they can be extracted from the 3GPP standard and its offered OpenAPI. Note that there has been no runtime anomaly detector for the 5G mobile network in the literature.

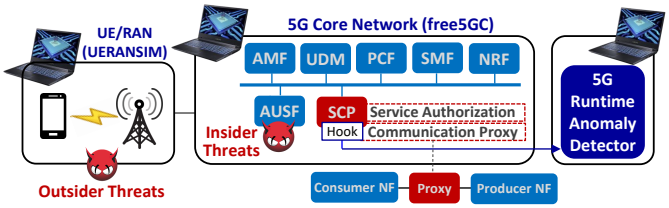


Fig. 1: 5G mobile network architecture with the 5G-RAD.

In the following, we first present the implementation of the SCP and the 5G-RAD platform, and then demonstrate their effectiveness with three potential security threats. The result shows that the 5G-RAD can successfully detect those threats by observing anomalous state machine or improper IE content.

II. 5G-RAD Platform

To develop the 5G-RAD platform, we build an experimental platform of the 5G mobile network architecture, as shown in Figure 1. It contains two major parts: (1) the 5G core network, which is built using the open-source `free5GC` [2]; (2) the UE and radio access network (RAN), which are simulated based on the open-source `UERANSIM` [3]. We then implement the SCP on the `free5GC` and equip it with a hook to duplicate control-plane messages to the 5G-RAD. At the platform, the module of control-plane message parsing and grouping processes the messages and transforms them from the HTTP/2 format to the JSON for the validation, as shown in Figure 2. The messages are further processed by the constraint and NF state machine validators to validate IEs and state machines, respectively. Any anomaly discovered from the validators will trigger an alert message.

For the validators, we build a state machine for each NF and constraints for each message in each procedure by analyzing

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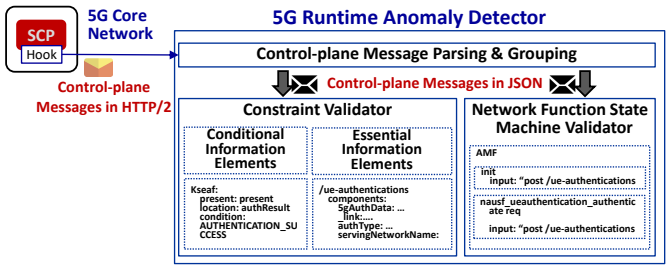


Fig. 2: 5G-RAD platform.

```
[UE 3][UDM] Validating constraints in HTTP request
[UE 3][UDM] Route          : POST /{supiOrSuci}/security-information/generate-auth-data
[UE 3][UDM] Done
[UE 3][AUSF][State Machine Error]: Expected Message:
    post /ue-authentications
[UE 3][AUSF][State Machine Error]: Received Message:
    post /{UE identifier}/security-information/generate-auth-data
-------------------------------------------------------
-                 DoS Attack detected                 -
-------------------------------------------------------
```

(a) Outsider threat: UE-originated DoS attack

```
[UE 0][AUSF] Validating constraints in HTTP response
[UE 0][AUSF] Route          : PUT /ue-authentications/{authCtxId}/5g-aka-confirmation
[UE 0][AUSF] Status         : 200
[UE 0][AUSF] Done
[UE 0][AUSF][State Machine Error]: Expected Message:
    post /{UE identifier}/security-information/generate-auth-data
[UE 0][AUSF][State Machine Error]: Received Message:
    put /ue-authentications/{UE identifier}/5g-aka-confirmation
-------------------------------------------------------
-        Authentication Bypass Attack detected        -
-------------------------------------------------------
```

(b) Insider threat: authentication bypass attack

Fig. 3: 5G-RAD demo results for outsider and insider threats.

the 3GPP OpenAPI and documents, and store them in the JSON format. We here mainly focus on the most important 5G procedure, 5G-AKA (Authentication and Key Agreement). The state machine can be built based on its procedure graph [1], whereas the essential and conditional IEs can be extracted from the 3GPP OpenAPI and documents. Notably, we consider both insider threats, which comes from inside the 5G core (e.g., a compromised NF), and outsider threats, which originates from outside the core (e.g., a malicious UE).

SCP. We implement two major functions of the SCP by following the 3GPP standard: service authorization and communication proxy. The former is developed based on the OAuth2.0 authorization framework to protect service access between NFs; i.e., the service consumer shall acquire a token to access a service from its producer. The latter's implementation is similar to a HTTP proxy that sits between NFs.

Control-plane message parsing/grouping module. It processes the control-plane messages for two tasks. First, it obtains IEs and the information (e.g., NF and messages names) related to per-NF state machines for each message, and store them in the JSON format. Second, it groups the messages based on a per-UE basis, so for each UE, we associate different IDs, i.e., SUCI and SUPI, based on a pair of HTTP request and response for their conversion process.

Constraint validator. It extracts the service data URI from each message to search for the corresponding constraints. It checks the IE presence (i.e., whether each IE shall appear) and the IE content (i.e., whether each IE's value is valid); notably, to determine the IE presence, the result of conditional event(s) shall be checked for the conditional IEs.

NF state machine validator. It keeps a state for each observed UE. Given a message, it uses the corresponding UE ID to search for the present state. If no state exists for the UE ID, the module will use the URI and associated NFs to discover which procedure the UE starts with. If an expected state exists, the validator will check whether it is consistent with the received message or not. The consistent result will guide the state machine to the next state; otherwise, an anomaly event will be generated.

III. Demonstration

We build the 5G testbed with `UERANSIM` and `free5GC` using two laptops and deploy the 5G-RAD platform using a laptop, as shown in Figure 1. To test the effectiveness of 5G-RAD, we implement one attack, UE-originated DoS, from outsider threats, and two attacks, authentication bypass and invalid message injection, from insider threats. As shown in the video at [4], the demonstration shows the effective operation of the SCP function and the 5G-RAD platform; in particular, the 5G-RAD can successfully detect those three attacks according to anomalous state machine or improper IE content.

Outsider threat. For the implementation of the UE-originated DoS in the `UERANSIM`, UEs keep initializing authentication procedures with different IDs but do not complete them. With a limited number of connection attempts, many incomplete authentication procedures may exhaust the resource. Figure 3a shows that the 5G-RAD can successfully detect such DoS attack. Specifically, the first column shows the assigned UE ID, which is associated with its SUCI and SUPI. In the log, the URI of the received message is '`.../generate-auth-data`'; the 5G-RAD detects that it is not expected in the state machine of AUSF, but an authentication response sent to AMF (i.e., '`/ue-authentications`') is anticipated.

Insider threat. The authentication bypass attack is assumed to be that the AUSF is compromised and allows UEs to bypass authentication to connect the 5G core; it requires the modification of both `free5GC` and `UERANSIM`. The AUSF and the UEs share a set of security context beforehand for the authentication so that the AUSF can skip retrieving authentication data from the UDM (Unified Data Management) while successfully authenticating the UEs. Figure 3b shows that this attack can be successfully detected by the 5G-RAD. In the AUSF state machine of the UE, the state machine validator expects a request message, '`.../generate-auth-data`', but receives another one, '`.../5g-aka-confirmation`'. For the invalid message injection, where the compromised AUSF skips a required IE (here, an authentication key, k_{seaf}), we modify the AUSF in `free5GC`. As shown in the demo video [4], the constraint validator can detect such attack.

References

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